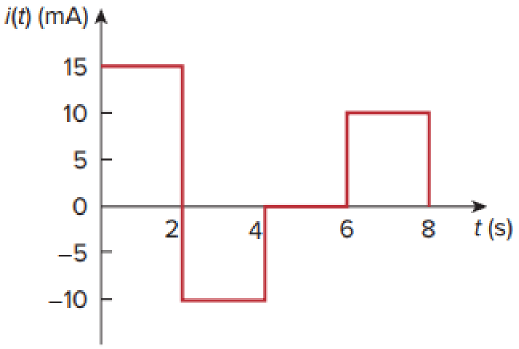
Chapter 6, Problem 11P

Problem

A 4-mF capacitor has the current waveform shown in Fig. 6.48. Assuming that $v(0)$ = 10 V, sketch the voltage waveform $v(t)$.

Fig. 6.48:



Step-by-step solution

Step 1 of 4

Refer to figure 6.48 for the waveform given in the textbook.

The initial voltage of the capacitor is,

$$v(0) = 10\text{ V}$$

From the waveform, the current values are,

$$i(t) = \begin{cases} 15 & 0 < t < 2\text{ s} \\ -10 & 2\text{ s} < t < 4\text{ s} \\ 0 & 4\text{ s} < t < 6\text{ s} \\ 15 & 6\text{ s} < t < 8\text{ s} \end{cases}$$

The expression for the voltage across the capacitor is,

$$v(t) = \frac{1}{C}\int i\,dt + v(0) \quad \text{...... (1)}$$

Calculate the voltage across the capacitor for $0 < t < 2s$.

Substitute 4×10^{-3} for *C*, 10 for $v(0)$, and 15×10^{-3} for *i* in equation (1).

$$\begin{aligned} v(t) &= \frac{1}{4 \times 10^{-3}}\int_0^t \left(15 \times 10^{-3}\right) dt + 10 \\ &= \frac{15 \times 10^{-3}}{4 \times 10^{-3}}\int_0^t 1 \times dt + 10 \\ &= 3.75 \times (t)\Big|_0^t + 10 \\ &= (3.75t + 10)\text{ V} \end{aligned}$$

Calculate the voltage at $t = 2\text{ s}$.

Substitute 2 for t.

$$\begin{aligned} v(2) &= 3.75(2) + 10 \\ &= 7.5 + 10 \\ &= 17.5\text{ V} \end{aligned}$$

Step 2 of 4

Calculate the voltage across the capacitor for $2 < t < 4s$.

Recall equation (1).

$$v(t) = \frac{1}{C}\int i dt + v(0)$$

Substitute 4×10^{-3} for *C*, 10 for $v(0)$, and 15×10^{-3} for *i*.

$$\begin{aligned} v(t) &= \frac{1}{4\times10^{-3}}\int_2^t \left(-10\times10^{-3}\right)dt + v(2) \\ &= (-2.5)t\big|_2^t + 17.5 \\ &= (-2.5t+5)+17.5 \\ &= -2.5t + 22.5 \text{ V} \end{aligned}$$

Calculate the voltage at $t = 4$ s.

$$\begin{aligned} v(4) &= -2.5(4) + 22.5 \\ &= -10 + 22.5 \\ &= 12.5 \text{ V} \end{aligned}$$

Step 3 of 4

Calculate the voltage across the capacitor for $4 < t < 6s$.Recall equation (1).

$$v(t) = \frac{1}{C}\int i dt + v(0)$$

Substitute 4×10^{-3} for *C*, 10 for $v(0)$, and 15×10^{-3} for *i*.

$$\begin{aligned} v(t) &= \frac{1}{4\times10^{-3}}\int_4^t (0)dt + v(4) \\ &= 0 + 12.5 \\ &= 12.5 \text{ V} \end{aligned}$$

Calculate the voltage across the capacitor for $6 < t < 8s$.

$$\begin{aligned} v(t) &= \frac{1}{4\times10^{-3}}\int_6^t \left(10\times10^{-3}\right)dt + v(6) \\ &= (2.5)t\big|_6^t + 12.5 \\ &= (2.5t-15)+12.5 \\ &= (2.5t-2.5) \text{ V} \end{aligned}$$

Therefore, the voltage across the capacitor is,

$$v(t) = \begin{cases} 10+3.75t \text{ V}, & 0 < t < 2 \text{ s} \\ 22.5-2.5t \text{ V}, & 2 < t < 4 \text{ s} \\ 12.5 \text{ V}, & 4 < t < 6 \text{ s} \\ \text{2.5t-2.5 V}, & 6 < t < 8 \text{ s} \end{cases}$$

Step 4 of 4

Draw the waveform of the voltage across the capacitor as shown in figure 1.

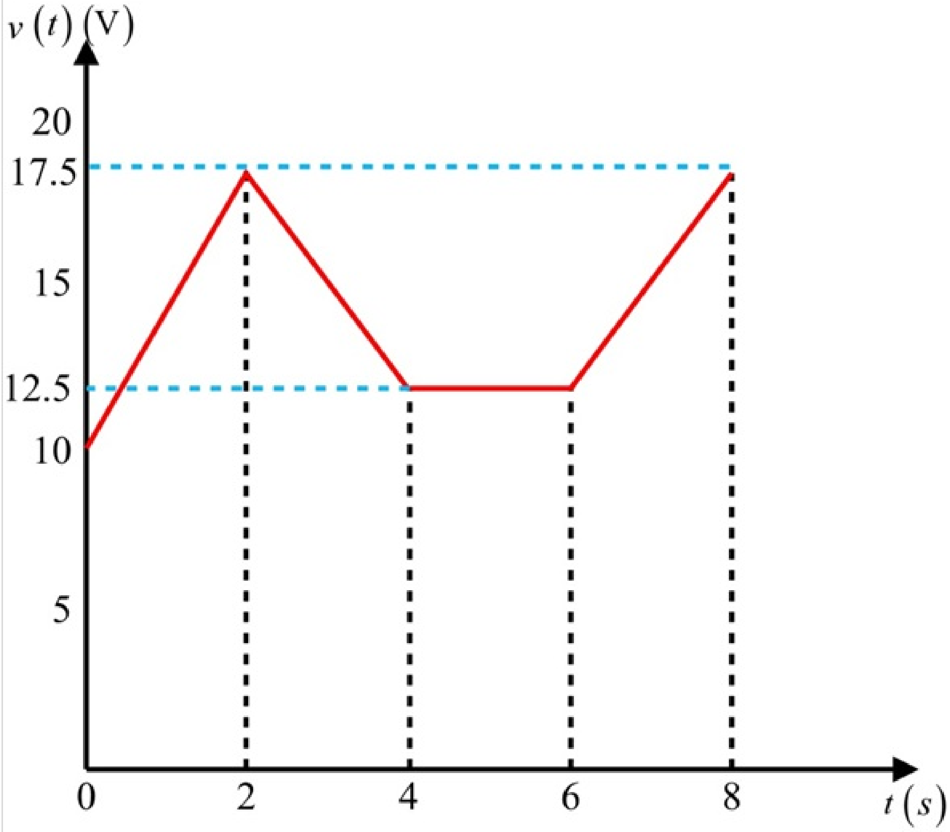


Figure 1